IBNU THYRA MAGANI

Secondary High School Student • Software Engineering & Financial Econometrics

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BINUS SCHOOL Semarang (Cambridge Assessment International Education) • Class of 2027 (Grade 11) • NISN: 0086859561

EDUCATION & ACADEMIC RECORD

BINUS SCHOOL Semarang — Cambridge Assessment International Education — Semarang, Indonesia • 2024 – Present

Secondary High School Student (Grade 11) • Combined Cumulative Academic Average (Grades 10–11): ***85.49 / 100***

Grade 10 Official Progress Report (AY 2024–2025, Class 10B)

Subject	Sem 1	Sem 2	Final	Grade
Bahasa Indonesia	92.59	93.87	93.23	A+
Physics	89.11	91.31	90.21	A
Mathematics	86.83	91.63	89.23	A
Sociology	89.93	89.15	89.54	A
Chemistry	91.78	86.05	88.92	A
Geography	90.51	85.43	87.97	A
Business Studies	87.24	87.92	87.58	A
Biology	90.63	80.83	85.73	A
English Language	81.23	81.50	81.37	A
Core Average	**88.87**	**87.52**	**88.20**	**Straight A**

Grade 11 Official Progress Summary (AY 2025–2026)

Subject	Sem 1	Sem 2	Final	Trajectory
Mathematics	85.0	92.0	88.5	+7.0
Physics	88.0	89.0	88.5	+1.0
Chemistry	85.0	86.0	85.5	+1.0
Biology	82.0	86.0	84.0	+4.0
PPKN (Civics)	89.0	93.0	91.0	+4.0
Bahasa Indonesia	85.0	85.0	85.0	0.0
Bahasa Jawa	82.0	89.0	85.5	+7.0
English Language	84.0	84.0	84.0	0.0
Islam Studies	78.0	74.0	76.0	-4.0
Physical Education	70.0	80.0	75.0	+10.0
Information Tech.	81.0	67.0	74.0	-14.0
Overall Average	**82.64**	**83.91**	**83.27**	**STEM: 86.6**

Transcripts authenticated via Certified True Copy signed by Principal Elsie Del Rosario Lallana Bait. Non-examinable electives completed with Grade A+ in School Band Ensemble. Attendance record: 95–100%.

STANDARDIZED TESTING & LANGUAGE CERTIFICATIONS

Digital SAT — Official College Board (Sep 12, 2026)

Total Score (81st Percentile Nationally)	**1260 / 1600**
Mathematics Section (90th Percentile)	**690 / 800**
• Algebra & Advanced Mathematics Content Domains	680–800 Band
• Problem-Solving, Data Analysis, Geometry & Trigonometry	610–670 Band
Reading & Writing Section (65th Percentile)	**570 / 800**
• Information and Ideas Content Domain	610–670 Band

IELTS Academic — Official Test Report (TRF: 26ID507816MAGI017A)

Overall Band Score (CEFR Level C1 — Proficient User)	**Band 7.0**
Reading	**7.5**
Listening	**7.0**
Speaking	**6.5**
Writing	**6.0**

Meets/exceeds direct English proficiency thresholds for English-track STEM university programs worldwide.

TECHNICAL COMPETENCIES & SYSTEMS STACK

Programming Languages & Web Architecture: Python 3.12, JavaScript (ES6+ / ES Modules), TypeScript, HTML5, CSS3, FastAPI (ASGI services), React 19, Vite, Tailwind CSS.

Numerical Computing & Data Engineering: NumPy (vectorized multidimensional array math), SciPy, Pandas, Plotly.js, Chart.js, HTML5 Canvas API.

Mathematical & Quantitative Methods: Geometric Brownian Motion (GBM), Itô's Lemma, Cox-Ross-Rubinstein (CRR) binomial lattice option pricing, Value-at-Risk (VaR), Conditional Value-at-Risk (CVaR / Expected Shortfall), Cornish-Fisher expansion for skewness/kurtosis adjustment, Parkinson (1980) High-Low & Garman-Klass (1980) realized volatility, Corwin-Schultz (2012) bid-ask spread proxy, Kapandji-Hansraj cervical biomechanical load curves.

DevOps, Privacy & Infrastructure: Git, GitHub Actions (CI/CD workflows), macOS launchd system daemons, Headless Chrome / Puppeteer (vector PDF compilation), MediaPipe Pose WebAssembly (client-side edge ML), Vercel, Supabase.

Engineering Methodology & AI Scaffolding: I formulate the governing physical equations and stochastic models from primary literature, leverage LLMs (Claude, DeepSeek) for initial API bindings and routine boilerplate, and personally execute numerical verification, boundary fixes, dependency elimination, and test suites.

TECHNICAL INTERESTS & RESEARCH FOCUS

- **Probability & Stochastic Processes:** Formulating continuous and discrete stochastic processes to quantify tail-risk boundaries and simulate forward asset distributions under market uncertainty.
- **Empirical Financial Econometrics:** Implementing realized volatility and liquidity proxies from raw order book and OHLCV data without relying on synthetic or idealized assumptions.
- **Automated Ingestion & Resilient Pipelines:** Designing self-healing ETL pipelines featuring network recovery daemons, cache-based idempotency, and automated document compilation.
- **Edge Computing & Client-Side Privacy:** Deploying computer vision and biometric evaluation pipelines entirely in-browser using WebAssembly to eliminate server-side media transmission.
- **Interactive Computational Simulations:** Translating formal academic syllabi (biochemistry, cellular kinetics) into deterministic state machines with hardware-accelerated procedural animations.

RESEARCH & SOFTWARE ENGINEERING PROJECTS

FutureAs — Quantitative Equity Valuation & Stochastic Engine [Live Demo • GitHub] 2025 – 2026

Python 3.12, FastAPI, NumPy, SciPy, Yahoo Finance API, React, Tailwind CSS, Plotly

- **Stochastic Engine:** Engineered a 10,000-path Geometric Brownian Motion (GBM) Monte Carlo simulation generating a 252-day forward matrix in under 80ms via vectorized NumPy cumulative log-return sums.
- **Discrete Option Lattice:** Programmed a Cox-Ross-Rubinstein (CRR) binomial lattice tree valuation model; personally derived and implemented the continuity correction to eliminate multi-step boundary convergence errors.
- **Downside Risk Analytics:** Formulated non-parametric 95% Value-at-Risk (VaR) and 95% Conditional Value-at-Risk (CVaR / Expected Shortfall).
- **Dependency Elimination:** Implemented the Abramowitz & Stegun rational polynomial approximation for inverse normal CDF, removing external SciPy runtime overhead and accelerating serverless cold-start execution.
- **Full-Stack Architecture:** Engineered 7 analytical modules (Valuation, Trend, Lump-Sum vs DCA, Probability Cones, Drawdowns, SEC Form 4/144 Insiders, Crash Radar) verified with a 15/15 passing automated test suite (`test_api.py`, `test_quant.py`).

QUANT.IDX Terminal — Indonesian Equity Econometric Terminal [Live Demo • GitHub] 2025 – 2026

Python 3.12, FastAPI, NumPy, Pandas, TradingView Pro Widget, Chart.js, Tailwind CSS

- **Extreme-Value Volatility:** Implemented 3 realized volatility estimators from econometric literature: Parkinson (1980) High-Low (~5x statistical efficiency over close-to-close metrics), Garman-Klass (1980) OHLC, and Rogers-Satchell (1991).
- **Non-Gaussian Tail Risk:** Modeled downside risk via the Cornish-Fisher expansion, adjusting VaR and CVaR for empirical skewness and excess kurtosis; implemented closed-form Chi-Square p-values for Jarque-Bera (1987) normality tests.
- **Liquidity Proxies:** Modeled empirical market liquidity using the Corwin-Schultz (2012) bid-ask spread proxy from 2-day high/low price ratios and the Amihud (2002) Illiquidity Ratio.
- **Full-Stack Interface:** Built a high-density financial interface embedding live TradingView Pro charts and automated DeepSeek econometric memorandums with deterministic offline fallbacks.

MediaSpine — In-Browser Clinical Posture Screener [Live Demo • GitHub] 2025 – 2026

Vanilla JavaScript (ES Modules), HTML5 Canvas, MediaPipe Pose (WebAssembly), DeepSeek V3, Supabase, Vercel

- **Zero-Cloud Privacy:** Runs MediaPipe BlazePose Lite locally via WebAssembly; frames are evaluated in memory and discarded immediately, guaranteeing 100% visual privacy with no cloud video uploads.
- **Automated Viewpoint Classifier:** Automatically separates Sagittal (profile) and Coronal (frontal) viewpoints based on dynamic horizontal bi-acromial (shoulder) landmark Euclidean distance thresholds.
- **Kinematics & Load Modeling:** Calculates 5 metrics (neck deviation angle, shoulder tilt, head tilt, 30-frame rolling stability variance, and 3-second hold stability gate). Modeled the Kapandji-Hansraj cervical spinal load curve (11 to 60 lbs).
- **Human Performance Engineering:** Personally implemented frame-drop throttling under GPU contention, circular buffers for landmark smoothing, and calibrated a deterministic 0–100 penalty scoring rubric.

Stock Intelligence Cloud — Autonomous Research & Dossier Pipeline [GitHub Repository] 2024 – 2026

Python 3.12, Puppeteer (Headless Chrome), GitHub Actions, macOS launchd, SMTP

- **Dual-Layer Redundancy:** Orchestrated an automated execution pipeline running locally at 09:00 WIB via macOS `launchd` daemons, paired with an automated GitHub Actions cloud cron workflow as autonomous fallback.
- **Idempotency & Resilience:** Enforced delivery idempotency via GitHub cache keys to prevent duplicate emails; built an automated network verification daemon with exponential backoff retry loops to prevent launchd crashes upon waking from sleep.
- **Vector PDF Publishing:** Automated vector PDF compilation using Headless Chrome for Testing with precision CSS print media pagination.

The Link Reaction — Interactive Cambridge 9700 Biological Simulation [Live Demo • GitHub] 2025

HTML5, CSS3 Keyframe Engine, Vanilla JavaScript (ES Modules), Procedural SVG

- **Biochemical State Engine:** Programmed an 8-stage finite state transition engine enforcing exact biological reaction steps: mitochondrial transport, decarboxylation (CO_2 release), dehydrogenation (NAD^+ -> $NADH$), and Coenzyme A ($CoA\text{-}SH$) sulfhydryl linkage.
- **Stoichiometric Conservation:** Enforced stoichiometric carbon conservation tracking 3C pyruvate splitting into 2C acetyl and 1C CO_2.
- **Zero-Dependency Performance:** Replaced heavy 3D WebGL libraries with procedural SVG transforms and CSS keyframes, ensuring zero external dependencies and 60 FPS performance across all low-power classroom devices.

HONORS, AWARDS & EXTRACURRICULAR LEADERSHIP

- **Silver Medal — Physics Engineering:** BISF (Bandung International Science Fair). Awarded for applied physics research, quantitative modeling, and experimental mechanical engineering analysis.
- **Cambridge Academic Distinction:** Straight A / A+ across all 9 examinable core subjects in Grade 10 (88.20% composite academic average).
- **Academic Resilience in Advanced Mathematics:** Demonstrated an upward performance trajectory in Grade 11 Cambridge Mathematics, advancing from 85.0 (Semester 1) to 92.0 (Semester 2).
- **School Band Elective:** Graded **A+**; active instrumentalist in school ensemble and public performances.
- **Athletics & Community Discipline:** Active member of BINUS School Badminton Club; maintained 95–100% attendance throughout high school career.